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Monsoon, 2024

B.Tech 1st Semester Examination

CHEMISTRY

Course Code : BCH23111T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION 1

1. Answer ALL questions: (10 × 1 = 10)

(i) Which of the following is infra-red (IR) active?

(a) HCl
(b) O_2
(c) N_2
(d) H_2

(ii) Which of the following is true for a good fuel?

(a) Low calorific value
(b) High ignition temperature
(c) Moderate ignition temperature
(d) None of the above

(iii) Internal energy is an

(a) Extensive property
(b) Intensive property
(c) Both intensive and extensive property
(d) None of the above properties

(iv) The lowest temperature at which an oil lubricant gives of vapours that ignites for a moment is called

(a) fire point
(b) flash point
(c) refractory point
(d) None of the above

(v) Refractoriness is measured by

(a)mechanical tests by UTM
(b) chemicals analysis
(c) thermogravimetric characterization
(d) pyrometric cone test

(vi) For an ideal gas, which of the following statement is correct

(a) $C_p = C_v$
(b) $C_p > C_v$
(c) $C_p < C_v$
(d) None of the above

(vii) Which among the following is paramagnetic?

(a) N_2

(b) O_2

(c) O_2^{2-}

(d) H_2

(viii) Waterline corrosion in steel tank is an example of

(a) Stress corrosion

(b) Differential aeration corrosion

(c) Pitting corrosion

(d) Differential metal corrosion

(ix) The corrosion can be prevented by

(a) alloying

(b) tinning

(c) galvanizing

(d) all of the above

(x) Bond order of O_2 molecule is

(a) 1

(b) 2

(c) 2.5

(d) 3

SECTION 2

Answer any one question from this section (Maximum 300 words for each) **(10)**

2. (a) (i) 10 moles of an ideal gas at the initial pressure of 1 atmosphere at 0 °C were expanded reversibly under isothermal conditions to a final pressure of 0.1 atmosphere. Calculate the work done by the gas, the change in internal energy and heat absorbed by the system (R=8.314 $JK^{-1}mol^{-1}$) (ii) Calculate the increase in entropy in the evaporation of one mole of water at 373 K. Latent heat of vaporization of water in 2.26 kJ/g (molar mass water 18). (6+4=10)

or

(b) (i) Determine the gross and net calorific value of a coal having the following compositions: carbon=85%, hydrogen=8%, Oxygen =0% ,sulphur=1%, nitrogen=2%, ash=4%, latent heat of steam=587cal/g. (5)

(ii) An electron in a one-dimensional box of width 2 Å undergoes a transition from the ground state (n=1) to the first excited state (n=2). Calculate the transition energy. (5)

SECTION 3

Answer any one question from this section (Maximum 300 words for each) **(10)**

3. (a) Explain the 12 principles of Green Chemistry with examples of each principle (10)

or

(b) (i) Determine the number of NMR signals in (4)

CH_3CH_2OH, $CH_2ClCHCl_2$, CH_3COCH_3, $CH_3CH_2CH_3$

(ii) Liquid fuel weighing 0.98 g and containing 90.1% C, 8% H have the following results in Bomb calorimeter experiment:

Amount of water taken in caloriemeter=1450 g

Water equivalent of calorimeter=450 g

Rise in temperature of water=1. 8 ^{0}C

If latent heat of steam is 587 cal/g, Determine the gross and net calorific value of fuel. (6)

SECTION 4

Answer any two question from this section (Maximum 300 words for each) **(2x5=10)**

4 (a) Distinguish between gross calorific value and net calorific value of a fuel. (5)

(b) List the different steps of setting and hardening of cement. (5)

(c) Distinguish between Galvanic corrosion and Pitting Corrosion. (5)

(d) List the consequences of corrosion. (5)

SECTION 5

Answer any one question from this section (Maximum 300 words for each) (10)

5 (a) (i) What is knocking? Give one example of an antiknocking agent. (3)

(ii) Enlist the factors influencing the rate of corrosion. (7)

or

(b) (i) Write the Schrodinger wave equation and explain the each term involved. (4)

(ii) What is Waterline corrosion? (3)

(iii) What is the role of gypsum in cement. (3)